

The subregular and submaximal p-cells

jt. V. Miemietz, D. Tubbenhauer.

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Today: – What are p-cells ?

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Today: – What are p-cells ?

– What do we know ?

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Today:
- What are p-cells ?
- What do we know ?
- Our description in some cases

I What are p-cells ?

Fix

. (W, S) a crystallographic Coxeter system

I What are p-cells ?

$s_1 s_2 s_1 s_2 = s_2 s_1 s_2 s_1$

Fix

. (W, S) a crystallographic Coxeter system

B_2 $\quad W = \langle s_1, s_2 \mid s_1^2 = s_2^2 = 1 = (s_1 s_2)^4 \rangle$

. $\mathbb{k}$ a Cartan realisation over $\mathbb{Z}$

e.g a root datum of a split reductive group over $\mathbb{Z}$

of the same type

$\mathbb{k} = \mathbb{Z} e_1 \oplus \mathbb{Z} e_2$

$\alpha_1 \qquad \alpha_2 \qquad \alpha_1(\alpha_2^\vee) = -2$

$\Delta = \{ e_1^* - e_2^*, e_2^* \}$ roots

$\alpha_2(\alpha_1^\vee) = -1$

$\Delta^\vee = \{ e_1 - e_2, 2e_2 \}$

$\alpha_1^\vee \qquad \alpha_2^\vee$

Hecke algebra

$H = H(W)$ over $\mathbb{Z}[v, v^{-1}]$ generated by $\{\delta_s\}_{s \in S}$

- $\delta_s^2 = (v^{-1} - v)\delta_s + 1$
- $\underbrace{\delta_s \delta_t \delta_s \cdots}_{m_{st}} = \underbrace{\delta_t \delta_s \delta_t \cdots}_{m_{st}}$

$(\delta_w)_{w \in W}$

$w = s_1 s_2$

$\delta_w = \delta_{s_1} \delta_{s_2}$

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Canonical basis $\{b_x\}_{x \in W}$

- $\overline{b_x} = b_x$

$\overline{(\cdot)} : v \mapsto v^{-1} \quad \delta_x \mapsto (\delta_{x^{-1}})^{-1}$

- $b_x = \delta_x + \sum_{y < x} v\mathbb{Z}_{\geq 0}[v]\,\delta_y$

$x = s_1 s_2 s_1$

$y = s_1 s_2$

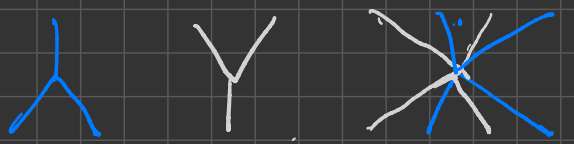
↳ Bruhat order

Diagrammatic category $\mathcal{H}_{BS}$

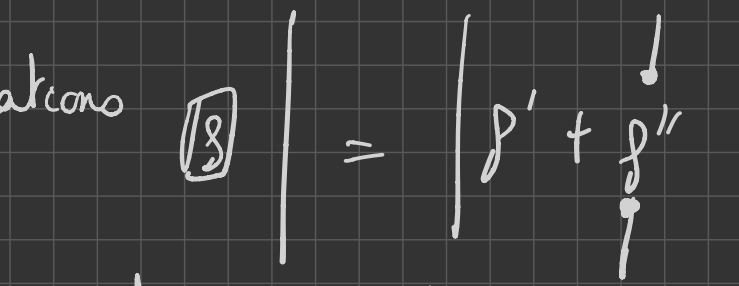
Fix K a complete local ring, $\mathfrak{h}_K = \mathfrak{h} \otimes_{\mathbb{Z}} K$, $R = \mathrm{Sym}(\mathfrak{h}_K^*)$.

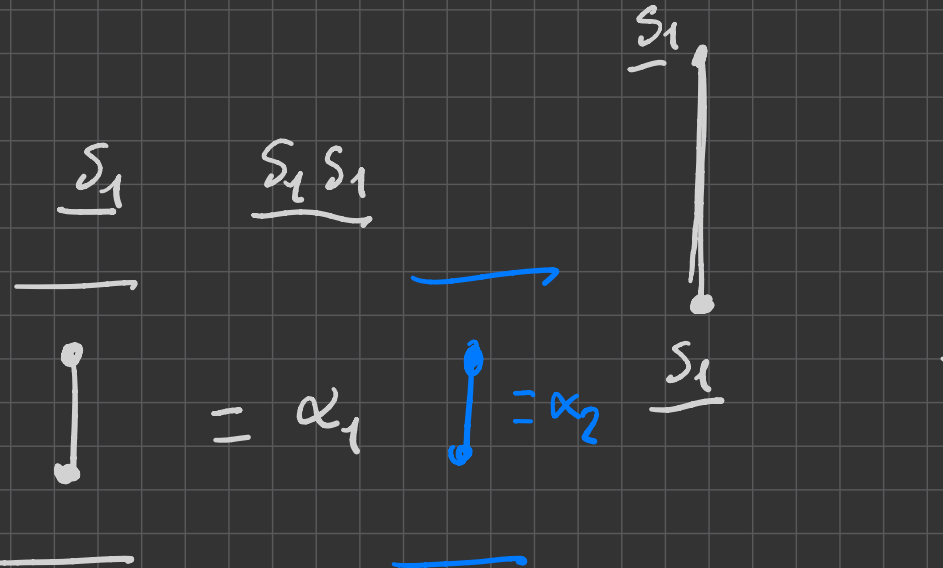
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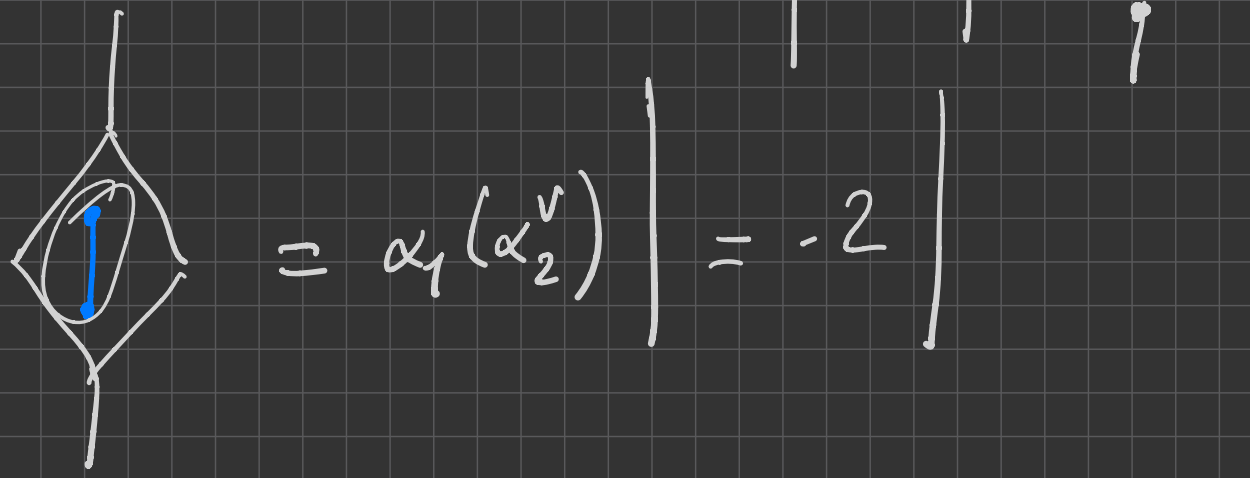
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- Objects : all expressions $\underline{w}$ of S
- Morphisms : S-graphs $\boxed{g}$ for $g \in R$ 

with relations







Diagrammatic Hecke category $\mathcal{H}$

$$\mathcal{H} = \mathrm{Kar}(\mathcal{H}_{BS}^{\oplus})$$

ie take direct sums and grading shifts, then direct summands

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Objects of the form $(\underline{w}, e)$

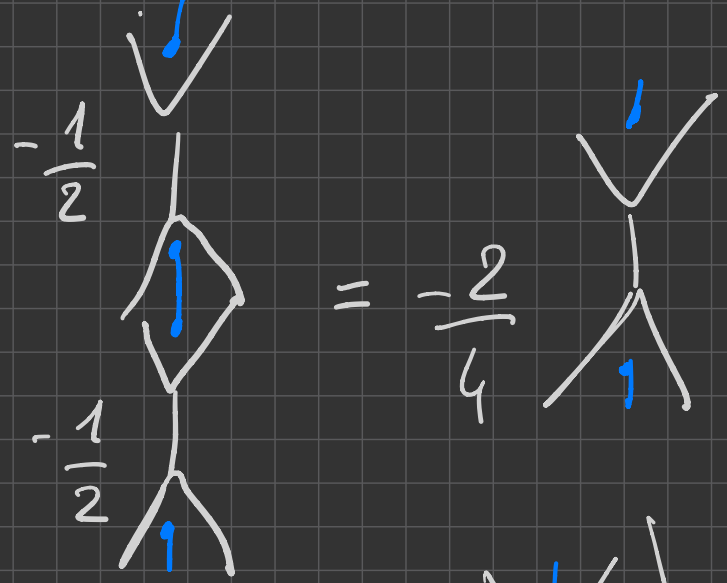
$e : \underline{w} \to \underline{w}$ idempotent

Morphisms $f : \underline{w} \to \underline{w}$ $foe = eof$.

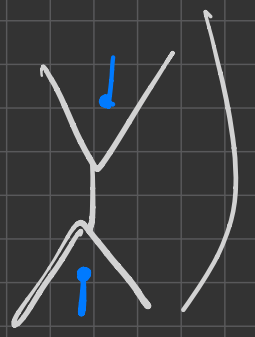
$(\underline{s_1}, |)$

$(\underline{s_1 s_2 s_1}, |||)$

char 0



$(\underline{s_1 s_2 s_1}, -\frac{1}{2}$



Theorem (Elias - Williamson '16)

1. $\mathcal{H}$ is Krull - Schmidt

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2. $\forall w \in W \quad \exists!$ B_w an indecomposable object in $\mathcal{H}$ s.t.
   $B_w \in (\underline{w}, id)$ and $\forall x \in W \; x \leq w$, $B_w \notin (\underline{x}, id)$

   ↑ Bruhat order

3. $\{B_w\}_{w \in W}$ complete set of representatives for indecomposable objects

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4. $\exists!$ isom of algebras $\mathbb{Z}[v, v^{-1}]$: $[\mathcal{H}^{\otimes}] \longrightarrow H$ $\quad s \in S$
   $[B_s] \longmapsto b_s$

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4. $\exists!$ isom of algebras $[\mathcal{H}^{\otimes}] \longrightarrow H$
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If K field $\rightarrow \operatorname{char} K = 0 \rightsquigarrow$ canonical basis
$\rightarrow \operatorname{char} K = p \rightsquigarrow$ p-canonical basis $\quad {}^{p}b_w = [{}^{p}B_w]$

Example of B_2

char 0

$$(\underline{s_1 s_2 s_1}, |||) = \underbrace{(\underline{s_1 s_2 s_1}, -\tfrac{1}{2}\ \text{Y})}_{[B_1]} + \underbrace{(\underline{s_1 s_2 s_1}, ||| + \tfrac{1}{2}\ \text{Y})}_{[B_{121}]}$$

$$[B_1 B_2 B_1] \cong [B_1] \oplus [B_{121}]$$

$$\downarrow$$

$$b_1 b_2 b_1 = b_1 + b_{121}$$

char 2 $(\underline{s_1 s_2 s_1}, |||)$ indecomposable

$$\downarrow {}^2B_{121}$$

$$\downarrow$$

$${}^p b_{121} = b_1 b_2 b_1 = b_1 + b_{121}$$

Cells

Let $F, G \in \mathcal{H}$ indecomposable

$F \leq_L G$ if $\exists H \in \mathcal{H}$ $G \subset HF$

$F \leq_R G$ if $\exists H \in \mathcal{H}$ $G \subset FH$

$F \leq_J G$ if $\exists H_1, H_2 \in \mathcal{H}$ $G \subset H_1 F H_2$

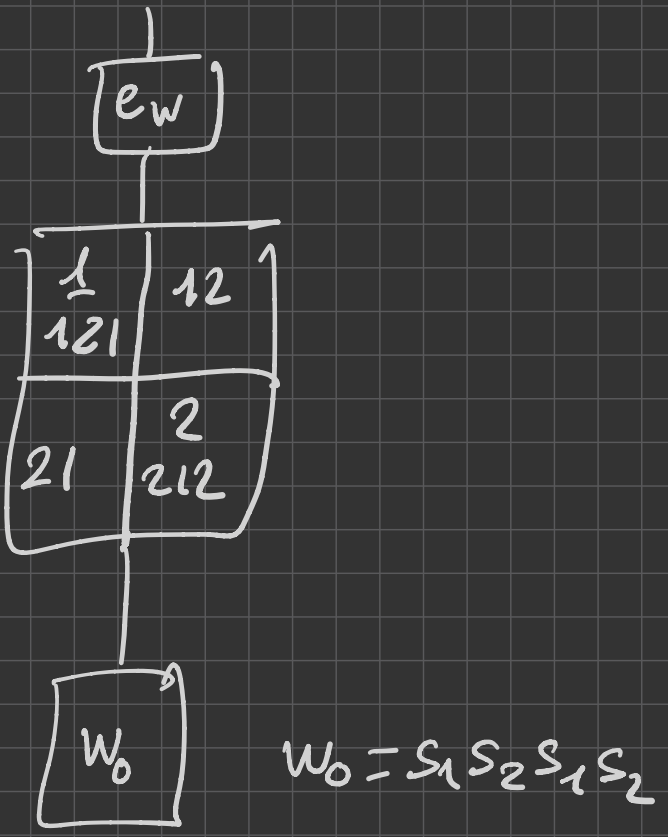
The equivalence relations are called left / right / two-sided (p)-cells

$B_2 B_1 = B_{21}$

$B_{21} \geq_L B_1$ clear

$B_1 B_{21} \cong B_1 \oplus B_{121}$

$B_1 \geq_L B_{21}$



$w_0 = s_1 s_2 s_1 s_2$

Aside / Why care?

- Geometry: 2-1 $\longleftrightarrow$ B-equivariant parity sheaves on G/B. over K

  0-cell $\longleftrightarrow$ Asymptotic Hecke category
  ↳ center gives character sheaves

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- Geometry: $\mathbb{Z}$-1 $\longleftrightarrow$ B-equivariant parity sheaves on G/B. over K

  0-cell $\longleftrightarrow$ Asymptotic Hecke category
  ↳ center gives character sheaves

- Representation theory: . reductive algebraic groups
  ↳ tilting modules

  char 0 . finite groups of Lie type (special characters, families ...).

Questions : 1) What is the p-canonical basis?

i.e. How to decompose ${}^{p}b_x$ into b_w $w \in W$?

2) What are the p-cells?

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i.e. How to decompose ${}^p b_x$ into b_w $w \in W$?

2) What are the p-cells?

$\rightsquigarrow$ There is an algorithm (Gibson, Jensen, Williamson '22)

But computer can't handle E_7, E_8, D_6.

Type A_n

$p = 2$ $n = 2$: ${}^{p}b_x = b_x \quad \forall x$

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$n = 7$: 38 out of 40320 elements with ${}^p b_x \neq b_x$

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Theorem (Williamson 2017) The size of T_n grows exponentially with n. $T_n := \{ p \mid \exists\, x \in A_n \;\; {}^p b_x \neq b_x \}$.

Type A_n

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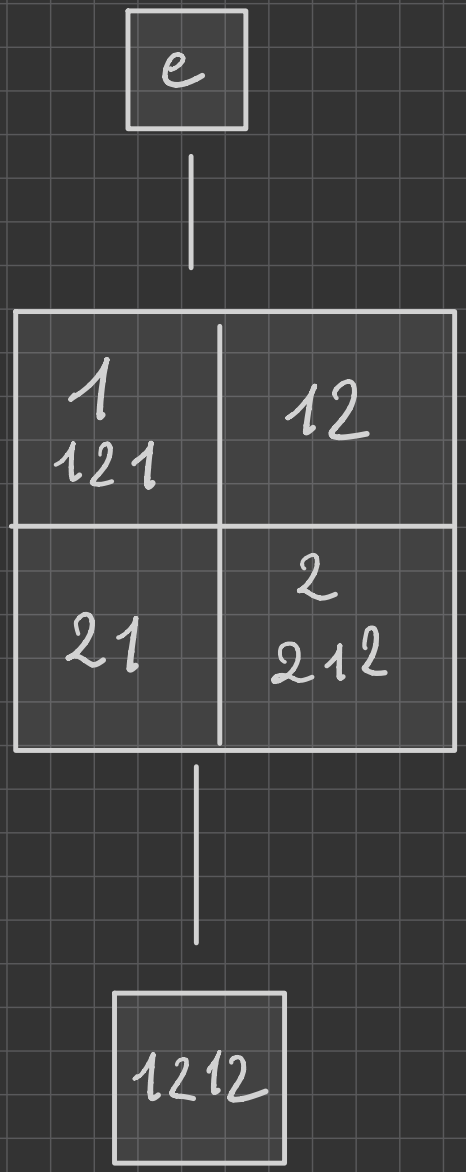
Theorem (Williamson 2017) The size of T_n grows exponentially with n. $T_n := \{p \mid \exists\, x \in A_n \ {}^p b_x \neq b_x\}$.

BUT

Theorem (Jensen 2020) $\forall$ prime p, the p-cells and 0-cells coincide in type A_n.

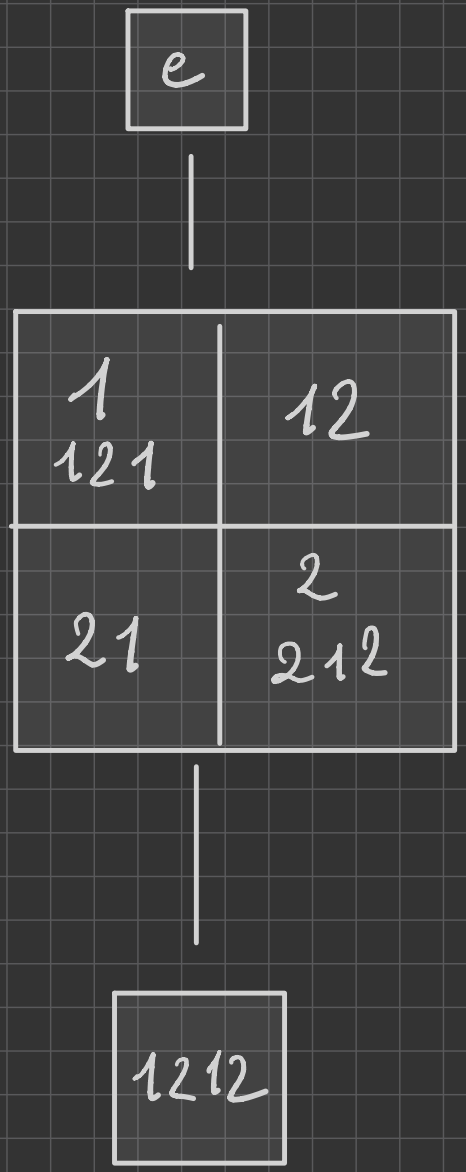
Type B_2

$p = 0$



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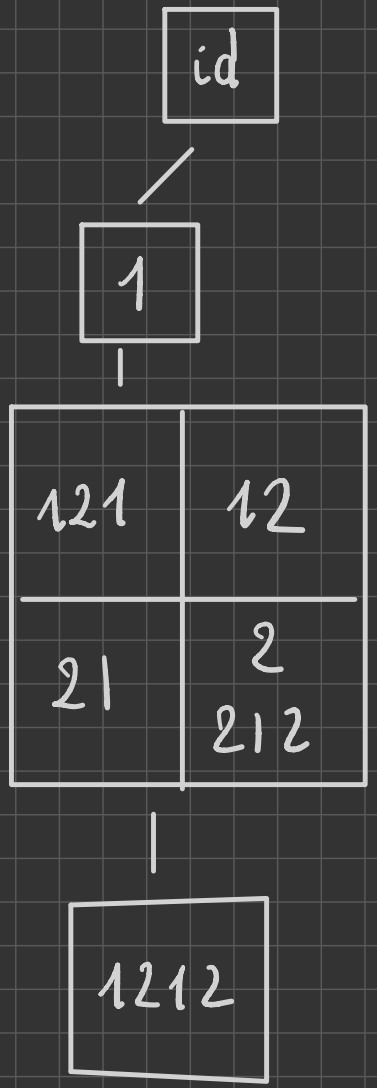
Some of type C_2

$p = 2$

Type B_2

$^{p}b_{121} = b_{121} + b_1$

$^{p}b_{212} = b_{212}$

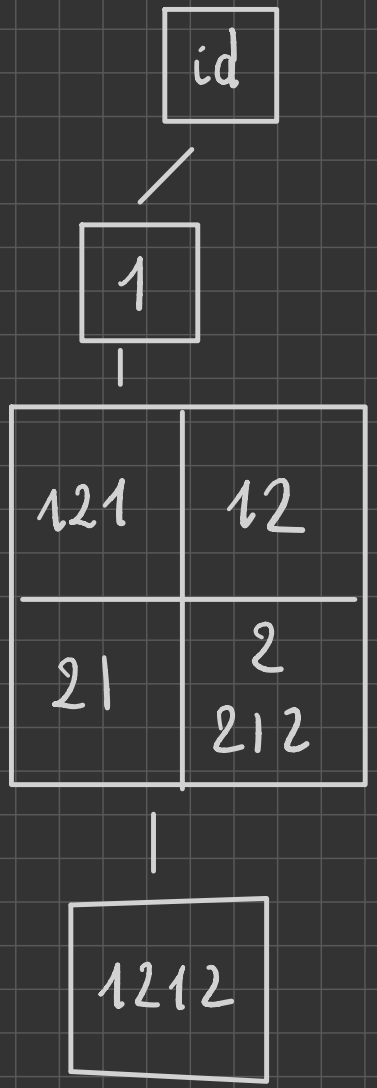


$p = 2$

Type B_2

$^p b_{121} = b_{121} + b_1$

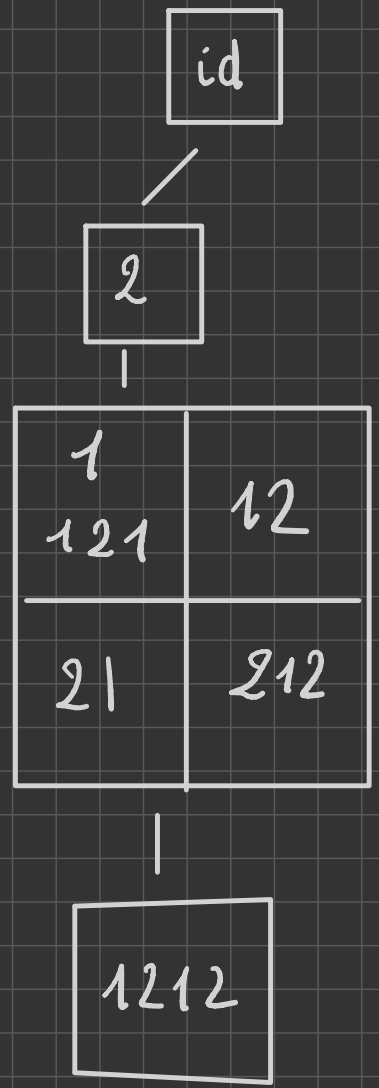
$^p b_{212} = b_{212}$



Type C_2

$^p b_{121} = b_{121}$

$^p b_{212} = b_{212} + b_2$



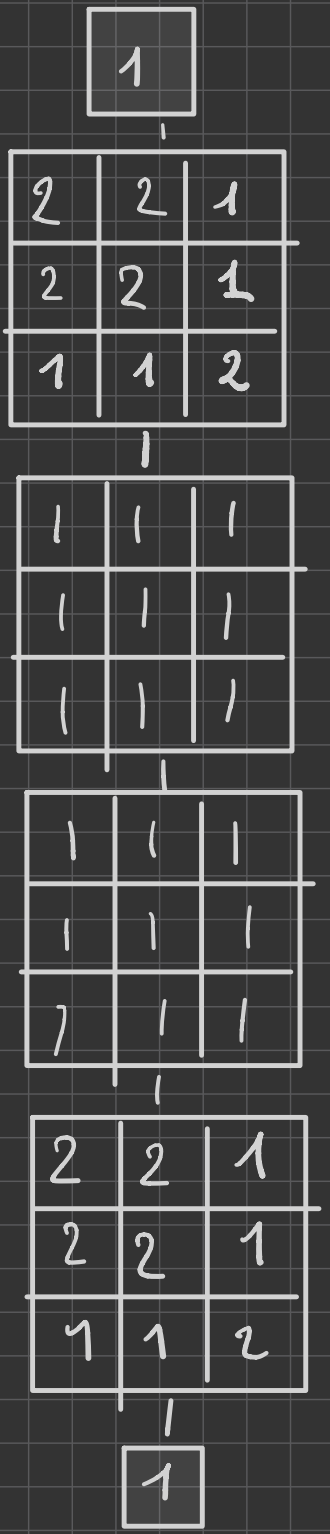
→ same "cell shape"

$p = 0$

Type B_3

Type C_3

same cell - shape



$p = 2$

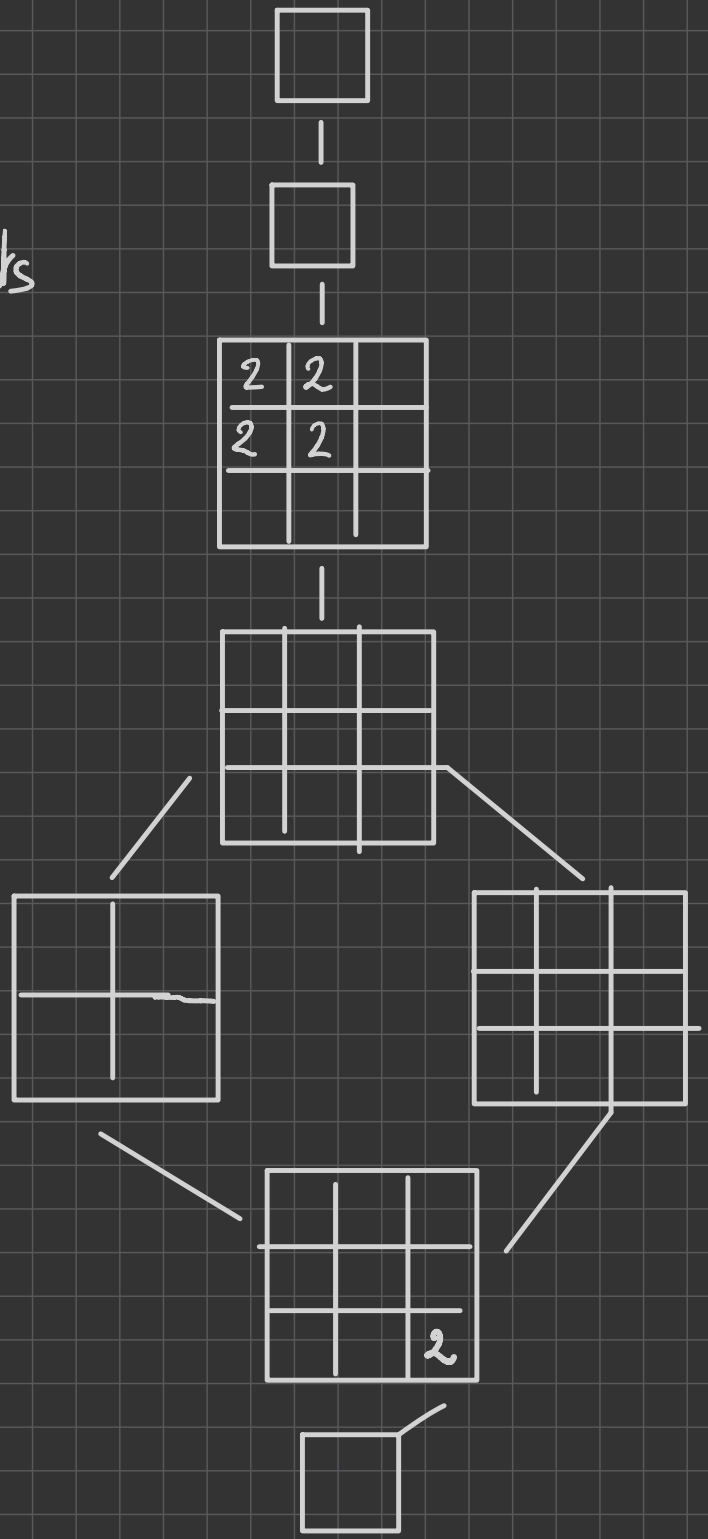
Type B_3

all coefficient are in $\mathbb{N}$.

$p = 2$

Type B_3

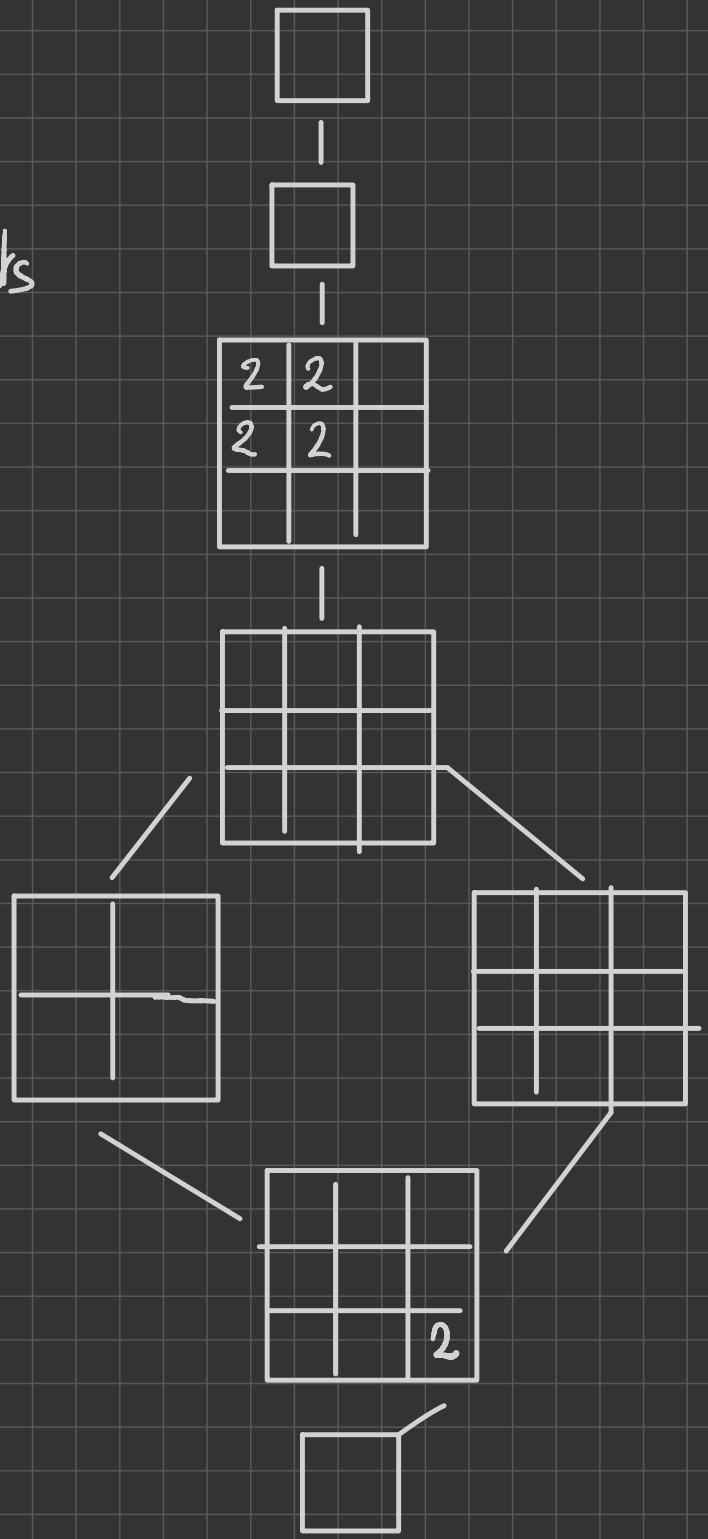
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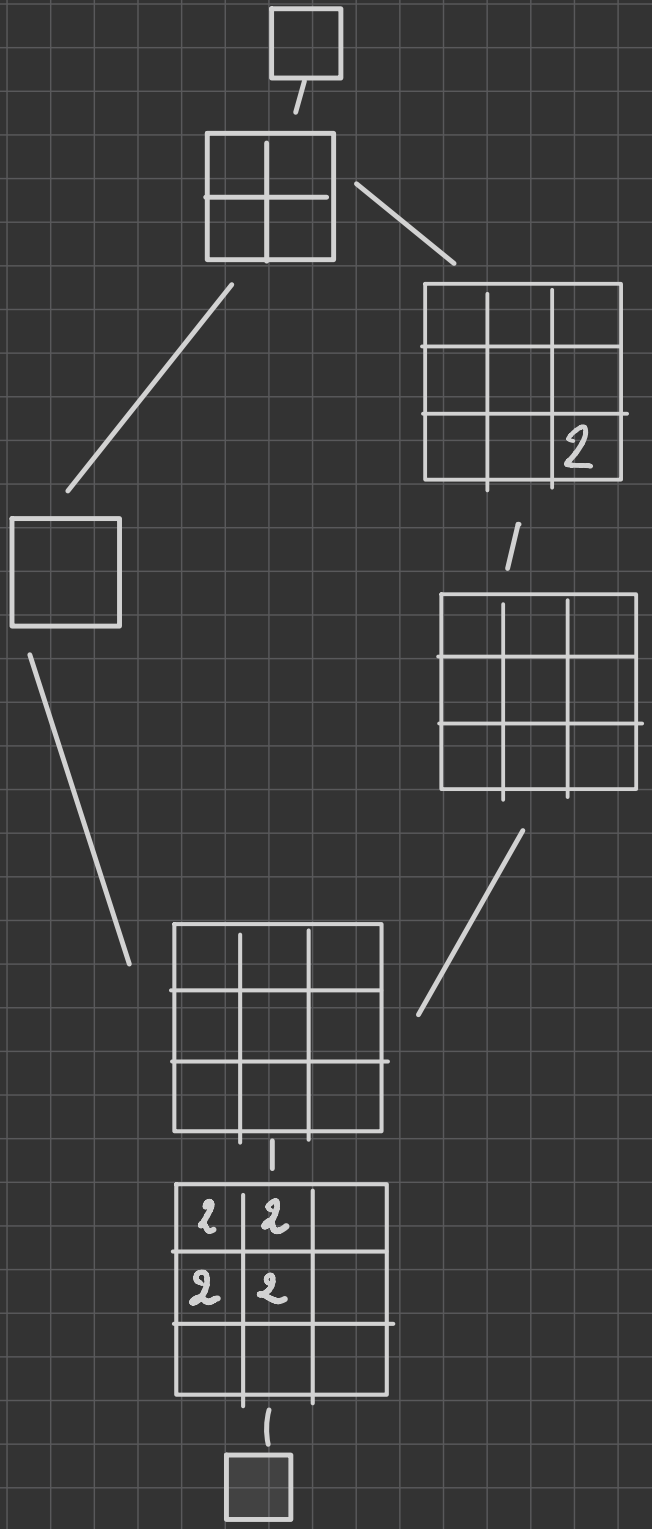
$p = 2$

Type B_3

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Type C_3



The subregular and submaximal cells

closest cell to id
↑
J

We focused on the elements in the subregular and the submaximal cell $\overline{J}$

↳ closest cell to w_0: longest element in W

The subregular and submaximal cells

We focused on the elements in the subregular J and the submaximal cell $\overline{J}$

	Elements of J	Elements of $\overline{J}$
type A_n and D_n	shortest paths from a to b $[a,b]$ $1 \le a,b \le n$	$\overline{x} := w_0 x$ for $x \in J$
type B_n and C_n	$[a,b]$ $1 \le a,b \le n$ and $[a,1,b]$ $1 \le a,b \le n$ $s_1 s_2 s_1$.	

$a < b$ $s_a s_{a+1} \cdots s_b = [a,b]$



$s_a s_{a-1} \cdots s_2 s_1 s_2 \cdots s_b$

Theorem (Miemietz - R. - Tubbenhauer) Let W be of classical type and J be the subregular 0-cell. Then, if $p = 2$

	J		$\overline{J}$	
	${}^{p}b_x \neq b_x$	p-cells	${}^{p}b_x \neq b_x$	p-cells
type A_n and D_n				
type B_n				
type C_n				

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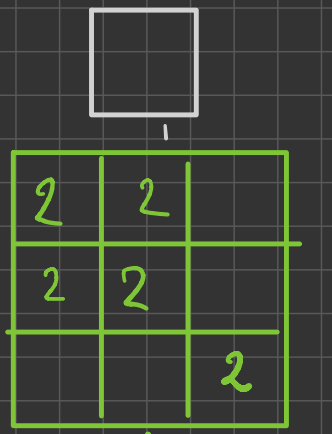
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type C_n	${}^p b_{[a,1,b]} = \sum_{\ell=1}^{\ell \leq a,b} {}^p b_{[a,\ell,b]}$	$J_1 \sqcup J_2$ $\shortparallel$ $\{[a,b]\}$	${}^p b_{\overline{1}} = b_{\overline{1}} + b_{\overline{121}}$ ${}^p b_{\overline{121}} = b_{\overline{121}} + ?$	

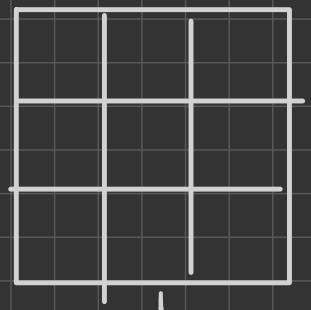
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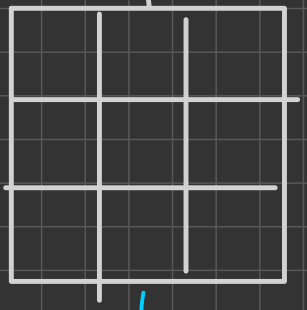
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type B_n	${}^p b_{121} = b_{121} + b_1$	$J_1 \sqcup \{s_1\}$	${}^p b_{\overline{[a,b]}} = b_{\overline{[a,b]}} + b_{\overline{[a,1,b]}}$	$J' \sqcup J''$ $| $ $\{\overline{[a,1,b]}\}$
type C_n	${}^p b_{[a,1,b]} = \sum_{\ell=1}^{\ell \leq a,b} b_{[a,\ell,b]}$	$J_1 \sqcup J_2$ $\|$ $\{[a,b]\}$	${}^p b_{\overline{1}} = b_{\overline{1}} + b_{\overline{121}}$ ${}^p b_{\overline{121}} = b_{\overline{121}} + ?$	$J' = \overline{J} \setminus \{\overline{121}\}$ and ?

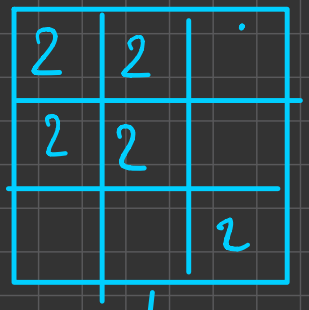
$p = 0$

type B_3 and C_3

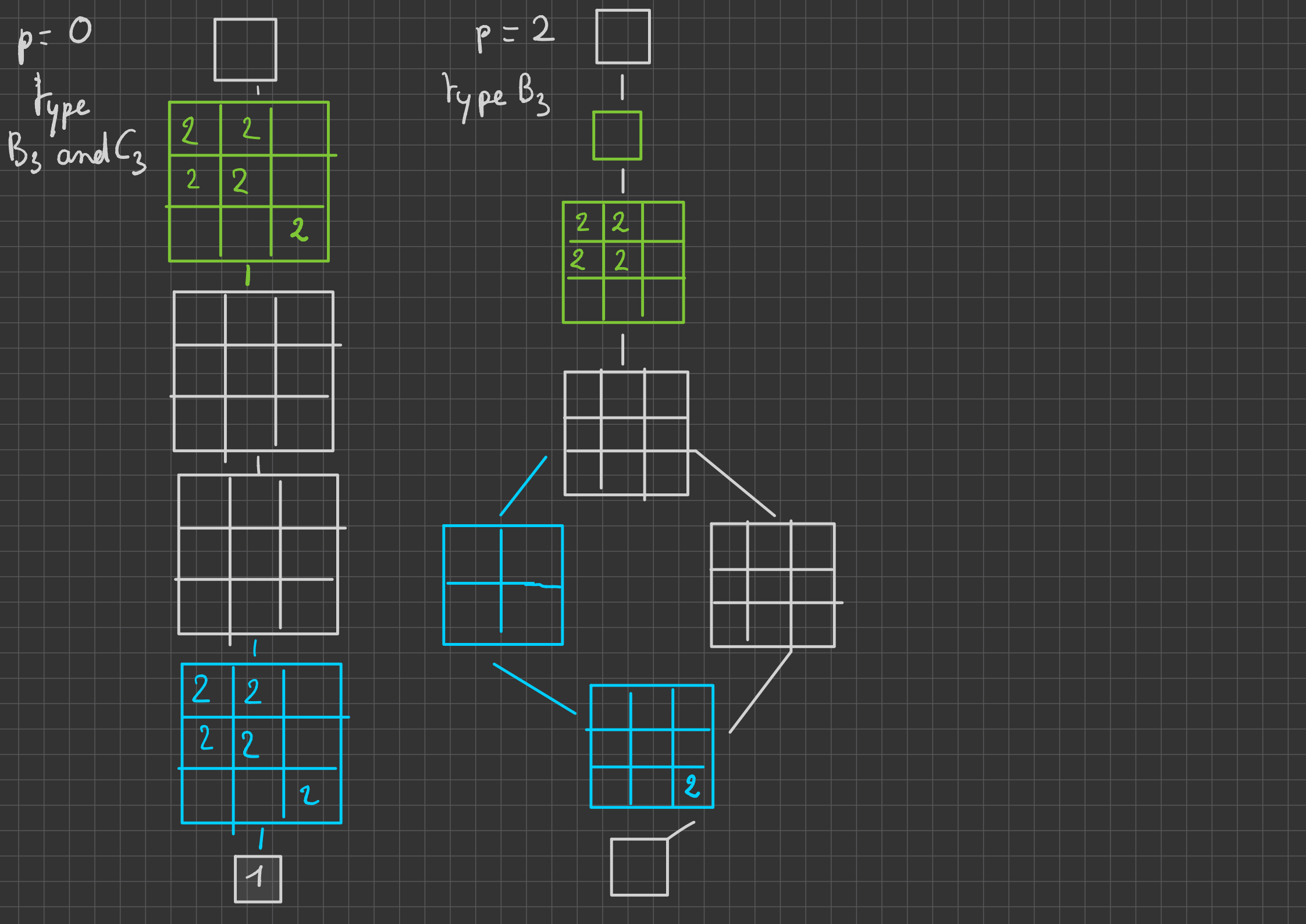


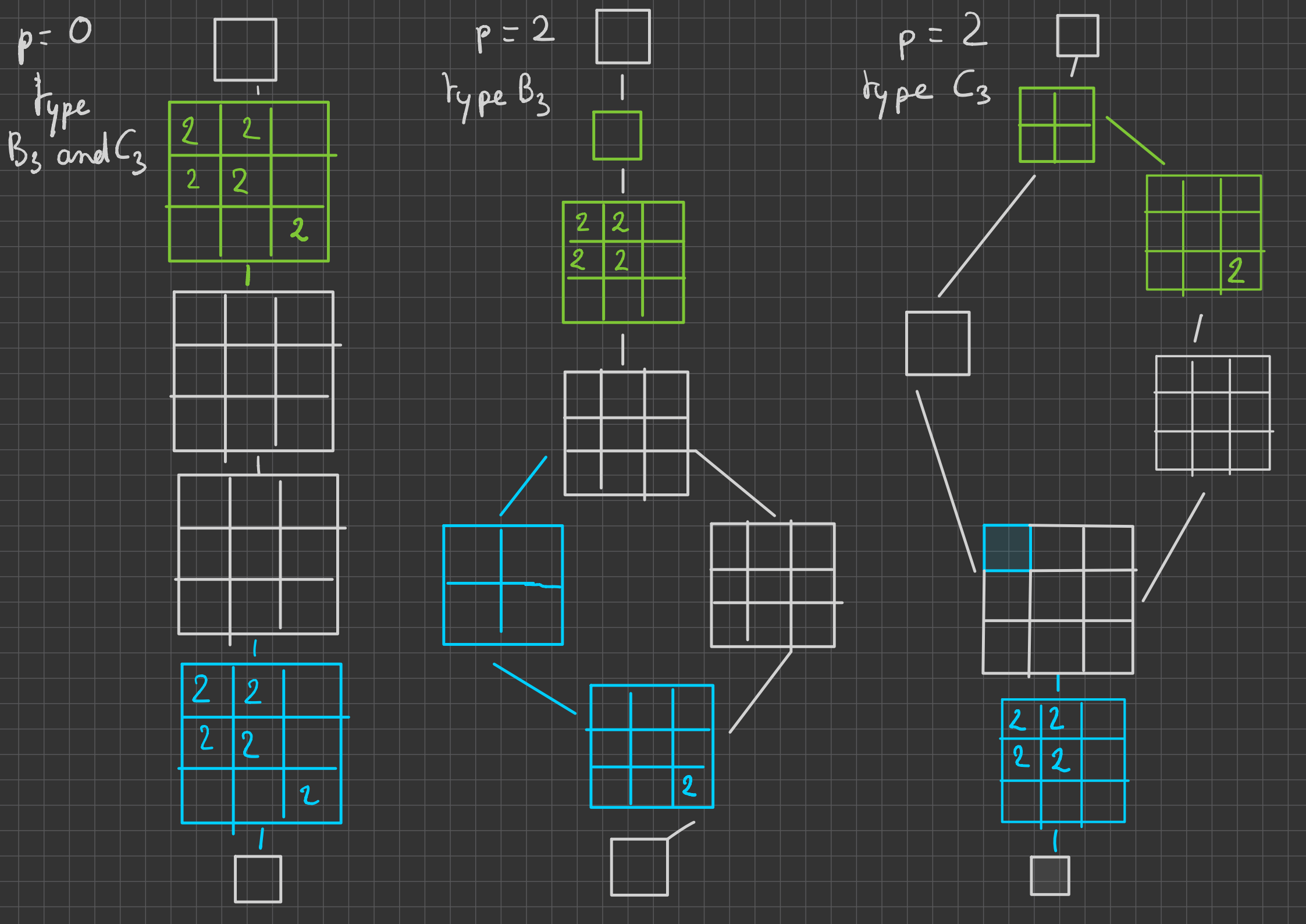









p = 0
type
B_3 and C_3
2 2
2 2
2
2 2
2 2
2 2
2 2
2
1
p = 2
type B_3
2
1


p = 0
type
B_3 and C_3
p = 2
type B_3
p = 2
type C_3

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- subregular very different than submaximal
- Lots of combinatorics
  - Step 1 : list possible $y \in W$ st b_y appears in $^p b_x$. $y \leq x$
    - ↳ subregular : constraint by Bruhat order
    - ↳ submaximal : constraint by left and right descent sets

$\mathcal{L}(x) \subseteq \mathcal{L}(y)$

$\mathcal{R}(x) \subseteq \mathcal{R}(y)$

$s_2 s_1 s_2$

$s_1 s_1 s_2 = s_2$

$s_1 \in \mathcal{L}(s_1 s_2)$

On the proof : • p-canonical basis, then p-cells

- subregular very different than submaximal
- Lots of combinatorics
  - Step 1 : list possible $y \in W$ st b_y appears in $^p b_x$.
    - ↳ subregular : constraint by Bruhat order
    - ↳ submaximal : constraint by left and right descent sets
  - Step 2 : check list using
    - ↳ induction
    - ↳ inverse
    - ↳ star operation $x \rightsquigarrow x^*$